

Mohammad Umar

Software Developer

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SUMMARY

Software Engineer who has designed, built, and shipped production full-stack systems, spanning React/Next.js frontends, Node.js/FastAPI backends, and AWS-deployed infrastructure. Delivered a live client platform end to end and built ML-driven applications integrating computer vision and Generative AI. Skilled in database design, authentication, testing, and deployment.

EXPERIENCE

Software Engineer | Full-Time | 01 Jul 2025 to 31 Jul 2026

Peer Networks Private Limited

- Developed and maintained 3+ client-facing and internal web applications using React.js, Next.js, Node.js, and Express.js, delivering features based on evolving business requirements
- Built 20+ REST API endpoints and reusable React components, integrating frontend applications with backend services and implementing authentication, validation, and error handling
- Worked across 2 database systems, PostgreSQL and MongoDB, supporting schema design, CRUD workflows, testing, debugging, and deployment on AWS (EC2, S3), along with application performance improvements

PROJECTS

DMcare360 | Live Client Project for DeckMount Electronics | Full-stack home healthcare platform

Next.js, Tailwind CSS, Supabase, Prisma, PostgreSQL

dmcare360.com

- Architected a production home healthcare platform with a full service catalog, reusable component library, Fuse.js-powered search, and a block-based blog CMS with Supabase Storage image uploads
- Built an AI-powered chat assistant using the Gemini API to help leads get instant answers about specific services
- Engineered a secure CRM admin with JWT + bcrypt authentication, role-based access control, and automated round-robin lead assignment via a Supabase cron job
- Implemented SEO-friendly, dynamically updated meta tags across all service pages to improve search visibility

FaceVitals | Browser-based vital signs estimation

React, FastAPI, Python, PyTorch

facevitals-tau.vercel.app | github.com/mohammad-umar-16/facevitals

- Built a browser-based tool that estimates heart rate, HRV, blood pressure, and SpO2 from live video, combining computer vision (478-point MediaPipe landmarking) with signal processing (CHROM, Welch PSD) and deep learning models (FacePhys.rlap, two-phase ResNet1D) on a FastAPI backend
- Diagnosed and fixed a core signal-extraction bug, then validated blood pressure estimates against 8,299 real-world cuff measurements, improving model reliability with ITA-based skin-tone correction and multi-ROI signal fusion

Realm | Real-time video calling with live multilingual translation
React, TypeScript, Node.js, Express, Socket.io, PostgreSQL (Neon), WebRTC
realm-vchat.vercel.app | github.com/mohammad-umar-16/realm

- Extended prior published research (IJSREM 2025) into a production video calling platform with live speech-to-text captioning and voice translation across 7 languages, building a WebRTC signaling layer with STUN/TURN failover and automatic reconnection handling
- Designed a multi-provider translation pipeline (DeepL, Sarvam AI, Gemini) with automatic fallback and a persistent shared cache to control API cost and stay within provider rate limits
- Implemented JWT/httpOnly-cookie authentication with email verification and real-time messaging via Socket.io using per-user rooms for instant delivery; deployed across Vercel, Render, and Neon Postgres

SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Frontend: React, figma, Next.js, Tailwind CSS, Vite, PyQt5, HTML, CSS

Backend & Testing: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, Prisma, SQLAlchemy, Playwright, Socket.io, WebRTC

Databases & Cloud: PostgreSQL, MongoDB, Supabase, Neon, AWS (EC2, S3), Vercel, Render, Git, Docker

AI & ML: PyTorch, Signal Processing, MediaPipe, Pandas, Gemini API, LLM Integration

EDUCATION

B.Tech, Computer Science and Engineering (Data Science) | 2021 to 2025

Indraprastha Engineering College, Dr. A. P. J. Abdul Kalam Technical University

CERTIFICATIONS

- Oracle Certified Foundations Associate | Oracle Data Platform 2025 (Oct 2025)
- Generative AI Foundations Certificate Program | upGrad & Microsoft (Oct 2025)
- Oracle Agentic AI Foundations Associate (1Z0-1157-26) (Aug 2026)
- Amazon Junior Software Developer | Coursera

ACHIEVEMENTS

- Qualified GATE 2026 in Computer Science and Information Technology (CSIT)
- Solved 100+ LeetCode problems, strengthening data structures and algorithms fundamentals

PUBLICATIONS

- Speech Translation Technology In Chatting And Video Conference Platform | IJSREM, Vol. 09, Issue 01 (Jan 2025), ISSN 2582-3930
ijsrem.com/download/speech-translation-technology-in-chatting-and-video-conference-platform
- Real Time Text and Video Communication System with Integrated Speech Translation | Accepted, ICCCA 2025

LANGUAGES

English, Hindi